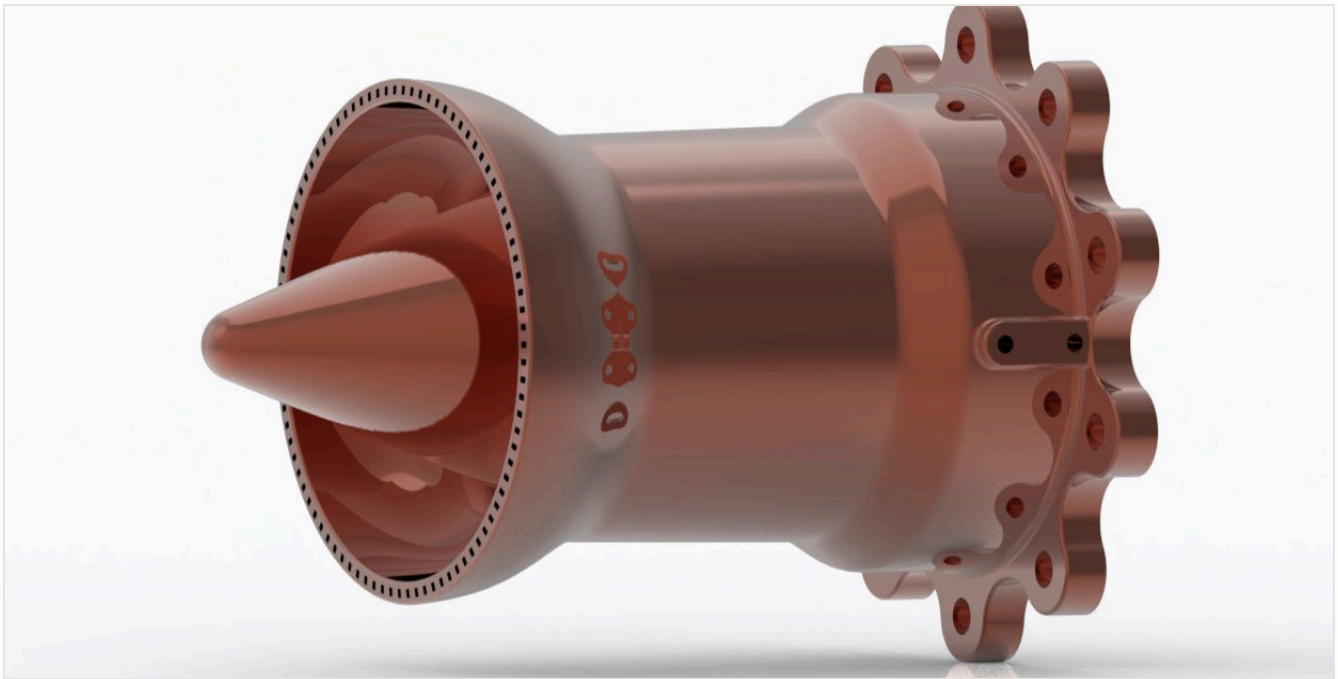


Joshua Dalal

I design, analyze, and build propulsion hardware.
Mechanical engineering student at the University of Pennsylvania (BSE + MSE) working across rocket propulsion, turbomachinery, and precision manufacturing, and co-founder of Project Wavefront, a nonprofit developing a rotating detonation rocket engine.



Project Wavefront Mark I rotating detonation rocket engine, 2 kN, GOX/GCH₄, additively manufactured

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Location

Philadelphia, PA

About

I'm a mechanical engineering student at the University of Pennsylvania pursuing a combined BSE + MSE in Mechanical Engineering and Applied Mechanics, with minors in Mathematics and History. My work focuses on propulsion and manufacturing: the analysis side and the hands-on build side in equal measure.

Experience

- May 2026 – Present, East Hartford, CT
Pratt & Whitney, Engine Dynamics & Loads Intern
Building data-driven models correlating aircraft structural characteristics with engine loads; improving reproducibility of NASTRAN fan blade-off analyses; experimentally characterizing squeeze-film damper performance in rotating machinery.
- Jun – Aug 2025
Hanwha Aerospace USA, Manufacturing Engineering Intern
Owned process flow design for the F135 Engine Core Upgrade rotor across 27 operations; PFMEA database development; fixture and tooling design driving a \$100k rate-increase clawback; qualified abrasive flow machining for a new part, cutting cost 35%.
- Jan – Jun 2025
Albacore Inc., Propulsion & Systems Engineer
Developed a UUV for the US Navy with a novel powertrain targeting 10× the range of existing models at 4× lower cost. Vehicle-level integration, subsystem prototyping and test; credited on a provisional patent filing.
- Aug 2024 – Present
UPenn MEAM, Teaching Assistant & Lab Assistant
Graduate machine design and manufacturing (IPD 5010): training students on 3-axis mills and CNC lathes, CNC programming, and design for manufacturing; shop maintenance and CAM guidance.
- 2024
GRASP Robotics Lab, Research Assistant
Designed a highly articulated robotic arm driven by a single base-mounted motor through a u-joint and bevel-gear transmission; prototyped a mechanical eyeball for an Air Force eye-tracking project reaching 0.3° accuracy at 600°/s.

Capabilities

Simulation & analysis

ANSYS Fluent CFD (combustion, species transport, transient) and FEA, MATLAB, Python. Coursework in compressible flow & combustion, aerodynamics, turbulence, heat & mass transfer, and machine design.

Design & CAD

Siemens NX, SolidWorks (CSWA), Onshape, surface modeling, KeyShot. ASME GD&T and tolerance stackup as a design tool, not an afterthought.

Manufacturing

CNC milling and turning (MasterCAM), manual machining, injection molding, laser cutting, additive manufacturing, DFM and process planning. Comfortable taking a part from drawing to inspection.

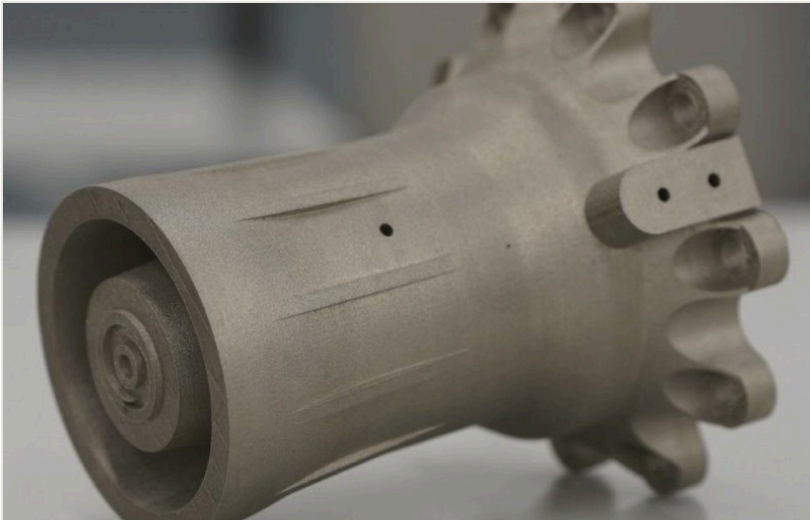
Education

University of Pennsylvania, BSE + MSE in Mechanical Engineering & Applied Mechanics; minors in Mathematics and History. 3.92 GPA. ASME and Wharton Undergraduate Aerospace Club.

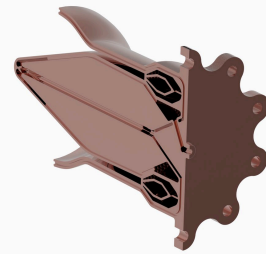
Rotating Detonation Rocket Engine

Co-founder and propulsion engineer at Project Wavefront, a nonprofit developing an affordable, additively manufactured GOX/GCH₄ rotating detonation rocket engine. The Mark I has cleared critical design review and printing; the test campaign is underway.

Thrust	P-chamber	Propellant	Isp	Wave freq.	Runtime
2 kN	21 bar	GOX / GCH ₄	255 s	13.5 kHz	>30 s



The Mark I demonstrator as printed



Half-section: annular combustor, cooling channels, aerospike

Why detonation

Conventional rocket engines burn propellant through subsonic deflagration. An RDRE instead sustains one or more detonation waves traveling continuously around an annular chamber. This pressure-gain combustion can improve thermodynamic efficiency on the order of 10–20%. The historical catch is cost: RDREs have demanded expensive infrastructure, complex manufacturing, and long development cycles. Project Wavefront targets that cost problem directly with a monolithic, additively manufactured methane/oxygen engine designed for minimal post-machining and rapid iteration.

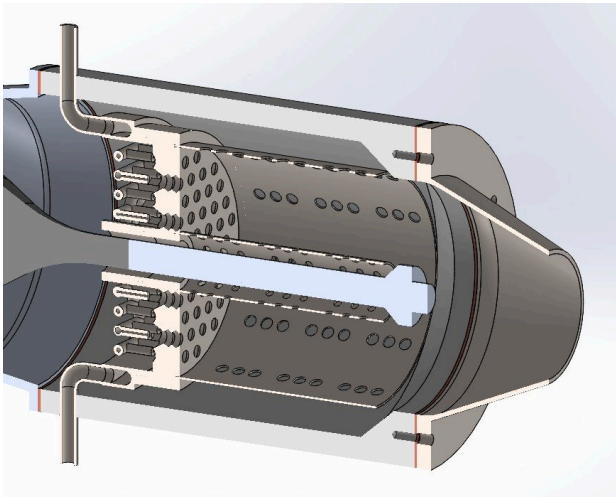
My work

As propulsion engineer I own much of the engine's fluid and combustion design: transient CFD of the impinging injectors, injector flow-distribution studies and geometry optimization, feed-line thermal and Joule-Thomson analysis, and blast-overpressure safety analysis for the test site. On the organizational side I co-founded the nonprofit itself: fundraising, sponsorships, vendor relationships across the feed-system supply chain, and the procurement program for the test stand now in buildout. Design freeze, critical design review, and demonstrator printing are complete, with cold flow and hot fire targeted for late summer 2026.

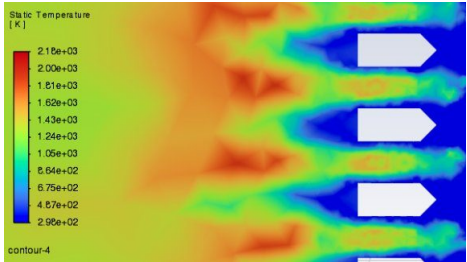
Hydrogen Turbofan Combustor

As propulsion team lead of Penn Jet Propulsion, I direct a team research project on hydrogen combustion through micromixing. Together we design and analyze the combustor, fuel manifold, and ignition system from first principles.

Fuel	Liner	Analysis	Role
Gaseous H ₂	304 SS	ANSYS Fluent / FEA	Team lead



Section view: perforated liner and fuel injector array



Combustor temperature field, ~2,180 K peak

The problem

Hydrogen's high flame temperature and wide flammability limits make it attractive for clean combustion but difficult to manage in compact hardware. The project's approach is micromixing: distributing fuel injection across many small jets to achieve stable, well-mixed combustion within the liner's material limits, using analysis-driven design rather than trial and error.

Analysis campaign

We built a CFD and FEA workflow around the combustor. Species-transport simulations map hydrogen mole fraction through the injector array to evaluate mixing quality and recirculation. Combustor-exit temperature contours size the dilution scheme and set liner material limits, and manifold simulations resolve pressure and temperature across the ring-and-stem injector network to verify uniform feed. Thermal-stress FEA of the 304 stainless liner closes the loop between gas-side temperatures and mechanical margin. The engine is built but has not yet been tested; the first test campaign is ahead.

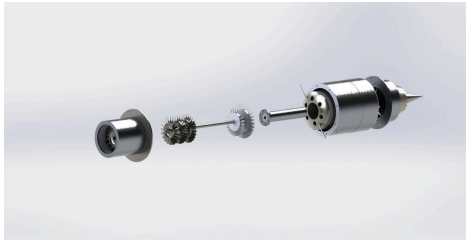
Axial Turbojet

Led the propulsion team on Penn's first undergraduate axial turbojet: aerodynamic profiling, structural integrity, and component optimization toward an 80,000 RPM, 35 lbf engine.

Target thrust	Spool speed	Architecture	Org
35 lbf	80,000 RPM	Axial	Penn Jet Propulsion



The assembled axial turbojet



Exploded CAD: spool, compressor and turbine stages, combustor, casings

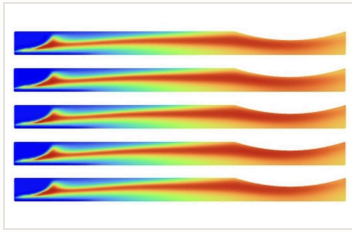
Engineering

Most student-built jet engines are centrifugal, which means simpler aerodynamics and more forgiving tolerances. We chose the axial route deliberately: it's the architecture full-scale propulsion engines use, and it requires solving blade aerodynamic profiling, rotordynamics at 80,000 RPM, thermal management around the combustor, and tolerances tight enough for stable operation at full speed. I led the team through thermodynamic cycle analysis, aerodynamic design of compressor and turbine stages, structural verification of rotating components, and combustion design.

Manufacturing

I also managed manufacturing planning and execution. The engine combines milled, turned, and additively manufactured components, all dimensioned with rigorous GD&T so that parts built by different students on different machines assemble into a balanced rotating machine.

CFD, Numerics & Mechatronics

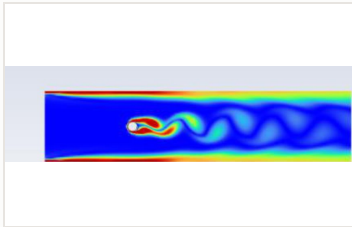


MEAM 6900, Spring 2026

Rocket Combustion Chamber CFD

GOX/GCH₄ combustion and supersonic nozzle flow in ANSYS Fluent with a species-transport model. Parametric sweeps of back pressure, O/F ratio, and injector swirl: rising back pressure suppressed nozzle choking and introduced normal shocks, while mixture-ratio sweeps shifted flame temperature with peak values near $MR \approx 4$. With Oren Minsk and Eitan Seitchik.

Report + slides available



MEAM 5360, Spring 2026

Flow Past a Circular Cylinder

2D incompressible flow at $Re = 0.1$ –1000 in ANSYS Fluent. A mesh-independence study settled on an 11k-element hybrid mesh with boundary-layer inflation; drag coefficients were benchmarked against empirical correlations, and transient runs captured the transition to periodic Kármán vortex shedding at $Re \geq 100$. With Jason de Gentile.

Report available

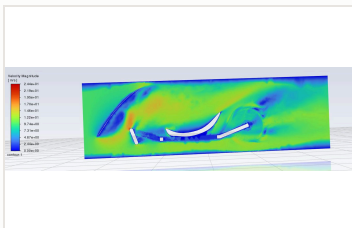


ENM 5220, Spring 2026

ADI Solver for Turing Patterns

A Peaceman–Rachford ADI solver for the Schnakenberg reaction-diffusion system, written from scratch in MATLAB: hand-derived discretization, Neumann boundary treatment at each half-step, and Thomas-algorithm tridiagonal solves. Reproduces the pattern-formation results of Madzvamuse & Maini (2007) with a convergence study.

Proposal available

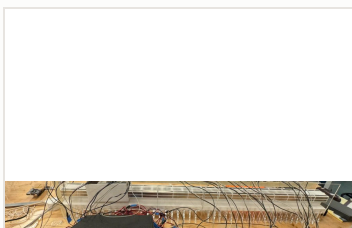


CFD Team Project, Spring 2025

Recumbent Bicycle Windshield

CFD study of windshield geometries for a recumbent bicycle, targeting rider comfort rather than pure drag reduction. Velocity-field analysis compared flat, interpolated, and curved shield profiles, focusing on flow conditions at the rider's head. With Annie Gao and Ari Weiner.

Slides available



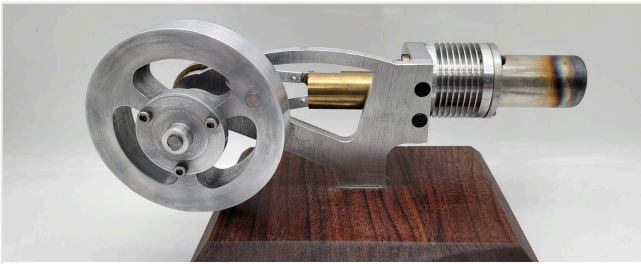
Mechatronics, 2025

Maglev Train

A magnetically levitated train with linear synchronous motor propulsion: N52 permanent-magnet levitation, air-core electromagnet coils driven by H-bridge controllers with Hall-effect commutation, and a two-part modular carriage with adjustable ballast to tune levitation height and lateral stability.

Report available

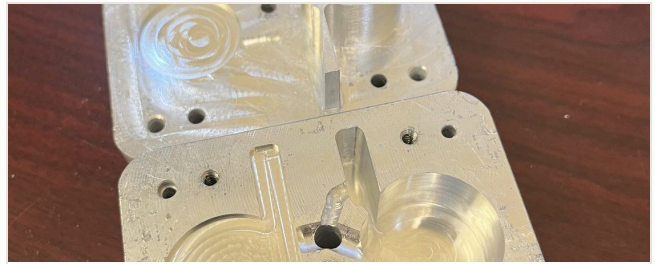
Precision Manufacturing



May 2024

Stirling Engine

18-part heat engine machined from aluminum, brass, stainless, and walnut, designed around a GD&T tolerance stackup.



Sept 2024, Team of 2

Injection Mold

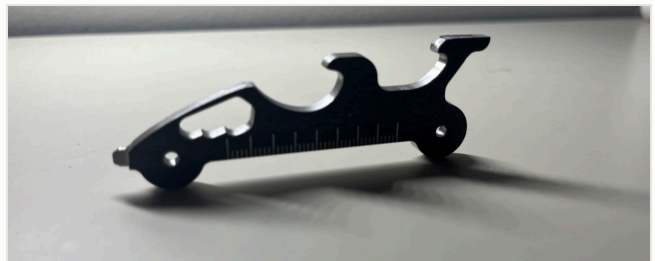
CNC-machined two-part mold for a working whistle: draft, parting lines, and shrinkage engineered in.



Oct 2024, Team of 4

Bauhaus Chess Set

Aluminum and brass chess set turned on a Haas TL-1 CNC lathe, with a walnut board.



2024

Racecar Multitool

Self-designed multitool with ruler, screwdriver, wrench, and bottle opener, fully toleranced to ASME GD&T.



2024, Team of 4

Music Box

Combined music box and cuckoo-clock mechanism, designed and manufactured end to end as a final project.



2024

SR-71 Display Model

CNC-machined aluminum display model of the SR-71 Blackbird, reduced to clean, machinable geometry.

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Project Wavefront

projectwavefront.org